A=LU factorization

Much earlier we looked at the idea of an elimination matrix as a matrix E that, when multiplied by another matrix A, performs a row operation in A. Now we want to use elimination matrices to rewrite A as the product of a lower triangular matrix L and an upper triangle matrix U, such that $A = LU$.

No row exchanges

When we perform an LU decomposition, or when we factor A into LU, we can't use row exchanges on the matrix A.

If row exchanges are required to put A into reduced row-echelon form (because A has a 0 in a pivot position), then we need to perform all those row exchanges first. Once any necessary row exchanges are complete, only then can we start the LU decomposition, which we'll perform on the new "row-exchanged" version of A.

How to change A into LU

Let's imagine that we're starting with a matrix A,

$$A = \begin{bmatrix} 5 & 1 & 3 \\ 4 & 2 & 5 \\ 8 & 6 & 4 \end{bmatrix}$$

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and we want to perform row operations on A until we've turned A into an upper triangular matrix. For instance, we would start to rewrite A as an upper triangular matrix by first changing the value in the second row and first column, $A_{2,1}$. To do that, we'll multiply A by an elimination matrix $E_{2,1}$.

$$E_{2,1}A = U$$

$$\begin{bmatrix} 1 & 0 & 0 \\ -\frac{4}{5} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 5 & 1 & 3 \\ 4 & 2 & 5 \\ 8 & 6 & 4 \end{bmatrix} = \begin{bmatrix} 5 & 1 & 3 \\ 0 & \frac{6}{5} & \frac{13}{5} \\ 8 & 6 & 4 \end{bmatrix}$$

The second row of the elimination matrix will replace the second row of A with $R_2 - (4/5)R_1$, which changes $A_{2,1}$ from a 4 to a 0. Then the goal is to keep applying elimination matrices to the left side of the equation, such that each elimination matrix performs a row operation on A that moves A one step closer to becoming the upper triangular matrix U.

Our next step would be to rewrite the 8 in $U_{3,1}$ as a 0,

$$E_{3,1}E_{2,1}A = U$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -\frac{8}{5} & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -\frac{4}{5} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 5 & 1 & 3 \\ 4 & 2 & 5 \\ 8 & 6 & 4 \end{bmatrix} = \begin{bmatrix} 5 & 1 & 3 \\ 0 & \frac{6}{5} & \frac{13}{5} \\ 0 & \frac{22}{5} & -\frac{4}{5} \end{bmatrix}$$

then to rewrite the $22/5$ in $U_{3,2}$ as a 0.

$$E_{3,2}E_{3,1}E_{2,1}A = U$$

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$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -\frac{22}{6} & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -\frac{8}{5} & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -\frac{4}{5} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 5 & 1 & 3 \\ 4 & 2 & 5 \\ 8 & 6 & 4 \end{bmatrix} = \begin{bmatrix} 5 & 1 & 3 \\ 0 & \frac{6}{5} & \frac{13}{5} \\ 0 & 0 & -\frac{31}{3} \end{bmatrix}$$

Now that U is an upper triangular matrix, we need to solve the equation $E_{3,2}E_{3,1}E_{2,1}A = U$ for A, which we can do by multiplying both sides by the inverse of each elimination matrix.

$$E_{3,2}^{-1}E_{3,2}E_{3,1}E_{2,1}A = E_{3,2}^{-1}U$$

$$E_{3,1}E_{2,1}A = E_{3,2}^{-1}U$$

$$E_{3,1}^{-1}E_{3,1}E_{2,1}A = E_{3,1}^{-1}E_{3,2}^{-1}U$$

$$E_{2,1}A = E_{3,1}^{-1}E_{3,2}^{-1}U$$

$$E_{2,1}^{-1}E_{2,1}A = E_{2,1}^{-1}E_{3,1}^{-1}E_{3,2}^{-1}U$$

$$A = E_{2,1}^{-1}E_{3,1}^{-1}E_{3,2}^{-1}U$$

In other words, to solve $E_{3,2}E_{3,1}E_{2,1}A = U$ for A, we move the entire group of elimination matrices to the other side of the equation, in front of U, reversing their order and inverting each one.

$$\begin{bmatrix} 5 & 1 & 3 \\ 4 & 2 & 5 \\ 8 & 6 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{4}{5} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -\frac{8}{5} & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -\frac{22}{6} & 1 \end{bmatrix}^{-1} \begin{bmatrix} 5 & 1 & 3 \\ 0 & \frac{6}{5} & \frac{13}{5} \\ 0 & 0 & -\frac{31}{3} \end{bmatrix}$$

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With the equation in this form, let's redefine the group of elimination matrices as L, so $L = E_{2,1}^{-1}E_{3,1}^{-1}E_{3,2}^{-1}$. If we say that L is the product of these inverted elimination matrices, then we can rewrite $A = E_{2,1}^{-1}E_{3,1}^{-1}E_{3,2}^{-1}U$ as $A = LU$, and all that's left to do is calculate L.

Inverting the elimination matrices to find L

To calculate L, we don't actually have to manually calculate the inverse of each elimination matrix. Instead, we know that the inverse of each elimination matrix can be found by changing the sign on the non-zero entry below the main diagonal.

In other words, given

$$\begin{bmatrix} 5 & 1 & 3 \\ 4 & 2 & 5 \\ 8 & 6 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{4}{5} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -\frac{8}{5} & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -\frac{22}{6} & 1 \end{bmatrix}^{-1} \begin{bmatrix} 5 & 1 & 3 \\ 0 & \frac{6}{5} & \frac{13}{5} \\ 0 & 0 & -\frac{31}{3} \end{bmatrix}$$

we can get the inverses by changing $-4/5$ to $4/5$, $-8/5$ to $8/5$, and $-22/6$ to $22/6$.

$$\begin{bmatrix} 5 & 1 & 3 \\ 4 & 2 & 5 \\ 8 & 6 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{4}{5} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \frac{8}{5} & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & \frac{22}{6} & 1 \end{bmatrix} \begin{bmatrix} 5 & 1 & 3 \\ 0 & \frac{6}{5} & \frac{13}{5} \\ 0 & 0 & -\frac{31}{3} \end{bmatrix}$$

Then, to calculate the product of the three inverted elimination matrices, we can simply combine the non-zero entries below the main diagonal into

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one matrix. So we'll take the $4/5$ from the first matrix, the $8/5$ from the second matrix, and the $22/6$ from the third matrix, and we'll get

$$\begin{bmatrix} 5 & 1 & 3 \\ 4 & 2 & 5 \\ 8 & 6 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{4}{5} & 1 & 0 \\ \frac{8}{5} & \frac{22}{6} & 1 \end{bmatrix} \begin{bmatrix} 5 & 1 & 3 \\ 0 & \frac{6}{5} & \frac{13}{5} \\ 0 & 0 & -\frac{31}{3} \end{bmatrix}$$

In this form, we see that L is a lower triangular matrix with all 1s along the main diagonal, and that U is an upper triangular matrix, which is exactly what we want. We've rewritten A as the product of the lower and upper triangular matrices, decomposing A into LU.

Another method

We can also accomplish this LU decomposition in a different way. We essentially set up the equation $A = IA$, where A is the matrix we're decomposing and I is the identity matrix, but with no entries filled in below the main diagonal.

This half empty identity matrix will become L while the matrix A on the right side of the equation becomes U. With this method, we'll collect row operations directly in L as we work to put A into reduced row-echelon form.

$$A = LU$$

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$$\begin{bmatrix} 5 & 1 & 3 \\ 4 & 2 & 5 \\ 8 & 6 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ & 1 & 0 \\ & & 1 \end{bmatrix} \begin{bmatrix} 5 & 1 & 3 \\ 4 & 2 & 5 \\ 8 & 6 & 4 \end{bmatrix}$$

We only need to zero-out the entries in U that are below the main diagonal. In this case, that means we're working on the 4, 8, and 6. To zero-out the 4 in $U_{2,1}$, we need to subtract $4/5$ of R_1 from R_2. Since that row operation is $R_2 - (4/5)R_1$, $4/5$ is the fraction we put into $L_{2,1}$.

$$\begin{bmatrix} 5 & 1 & 3 \\ 4 & 2 & 5 \\ 8 & 6 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{4}{5} & 1 & 0 \\ & & 1 \end{bmatrix} \begin{bmatrix} 5 & 1 & 3 \\ 0 & \frac{6}{5} & \frac{13}{5} \\ 8 & 6 & 4 \end{bmatrix}$$

To zero-out the 8 in $U_{3,1}$, we need to subtract $8/5$ of R_1 from R_3. Since the row operation is $R_3 - (8/5)R_1$, $8/5$ is the fraction we put into $L_{3,1}$.

$$\begin{bmatrix} 5 & 1 & 3 \\ 4 & 2 & 5 \\ 8 & 6 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{4}{5} & 1 & 0 \\ \frac{8}{5} & & 1 \end{bmatrix} \begin{bmatrix} 5 & 1 & 3 \\ 0 & \frac{6}{5} & \frac{13}{5} \\ 8 & 6 & 4 \end{bmatrix}$$

Finally, to zero-out the $22/5$ in $U_{3,2}$, we need to subtract $22/6$ of R_2 from R_3. Since the row operation is $R_3 - (22/6)R_2$, $22/6$ is the fraction we put into $L_{3,2}$.

$$\begin{bmatrix} 5 & 1 & 3 \\ 4 & 2 & 5 \\ 8 & 6 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{4}{5} & 1 & 0 \\ \frac{8}{5} & \frac{22}{6} & 1 \end{bmatrix} \begin{bmatrix} 5 & 1 & 3 \\ 0 & \frac{6}{5} & \frac{13}{5} \\ 0 & 0 & -\frac{31}{3} \end{bmatrix}$$

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With this method, it helps to always write the row operations with subtraction as $R_i - CR_j$. This way, the constant C is the value that goes directly into L. If we write the row operation with addition as $R_i + CR_j$, then $-C$ will be the value we put into L.

Let's try an example with a larger matrix.

Example

Rewrite the matrix A in factored LU form.

$$A = \begin{bmatrix} 1 & 3 & 4 & 0 \\ 9 & 9 & 0 & 7 \\ 7 & 7 & 6 & 5 \\ 0 & 3 & 0 & 2 \end{bmatrix}$$

Let's find the LU decomposition using both methods we outlined earlier. We'll start by finding the elimination matrices we need to apply to A in order to put it into reduced row-echelon form.

$$E_{2,1}A = U$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ -9 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 4 & 0 \\ 9 & 9 & 0 & 7 \\ 7 & 7 & 6 & 5 \\ 0 & 3 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 4 & 0 \\ 0 & -18 & -36 & 7 \\ 7 & 7 & 6 & 5 \\ 0 & 3 & 0 & 2 \end{bmatrix}$$

$$E_{3,1}E_{2,1}A = U$$

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$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -7 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ -9 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 4 & 0 \\ 9 & 9 & 0 & 7 \\ 7 & 7 & 6 & 5 \\ 0 & 3 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 4 & 0 \\ 0 & -18 & -36 & 7 \\ 0 & -14 & -22 & 5 \\ 0 & 3 & 0 & 2 \end{bmatrix}$$

$$E_{3,2}E_{3,1}E_{2,1}A = U$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -\frac{14}{18} & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -7 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ -9 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 4 & 0 \\ 9 & 9 & 0 & 7 \\ 7 & 7 & 6 & 5 \\ 0 & 3 & 0 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 3 & 4 & 0 \\ 0 & -18 & -36 & 7 \\ 0 & 0 & 6 & -\frac{4}{9} \\ 0 & 3 & 0 & 2 \end{bmatrix}$$

$$E_{4,2}E_{3,2}E_{3,1}E_{2,1}A = U$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & \frac{3}{18} & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -\frac{14}{18} & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -7 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ -9 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & 4 & 0 \\ 9 & 9 & 0 & 7 \\ 7 & 7 & 6 & 5 \\ 0 & 3 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 4 & 0 \\ 0 & -18 & -36 & 7 \\ 0 & 0 & 6 & -\frac{4}{9} \\ 0 & 0 & -6 & \frac{19}{6} \end{bmatrix}$$

$$E_{4,3}E_{4,2}E_{3,2}E_{3,1}E_{2,1}A = U$$

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$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & \frac{3}{18} & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -\frac{14}{18} & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -7 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ -9 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 4 & 0 \\ 9 & 9 & 0 & 7 \\ 7 & 7 & 6 & 5 \\ 0 & 3 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 4 & 0 \\ 0 & -18 & -36 & 7 \\ 0 & 0 & 6 & -\frac{4}{9} \\ 0 & 0 & 0 & \frac{49}{18} \end{bmatrix}$$

To solve this equation for A, we need to move all the elimination matrices to the right side, reversing the order and inverting each one.

$$\begin{bmatrix} 1 & 3 & 4 & 0 \\ 9 & 9 & 0 & 7 \\ 7 & 7 & 6 & 5 \\ 0 & 3 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -9 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -7 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -\frac{14}{18} & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}^{-1}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & \frac{3}{18} & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 3 & 4 & 0 \\ 0 & -18 & -36 & 7 \\ 0 & 0 & 6 & -\frac{4}{9} \\ 0 & 0 & 0 & \frac{49}{18} \end{bmatrix}$$

To invert each elimination matrix, we'll change the sign on the non-zero entry below the main diagonal,

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$$\begin{bmatrix} 1 & 3 & 4 & 0 \\ 9 & 9 & 0 & 7 \\ 7 & 7 & 6 & 5 \\ 0 & 3 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 9 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 7 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & \frac{7}{9} & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -\frac{1}{6} & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 4 & 0 \\ 0 & -18 & -36 & 7 \\ 0 & 0 & 6 & -\frac{4}{9} \\ 0 & 0 & 0 & \frac{49}{18} \end{bmatrix}$$

then we'll consolidate all the non-zero entries into one matrix, L.

$$\begin{bmatrix} 1 & 3 & 4 & 0 \\ 9 & 9 & 0 & 7 \\ 7 & 7 & 6 & 5 \\ 0 & 3 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 9 & 1 & 0 & 0 \\ 7 & \frac{7}{9} & 1 & 0 \\ 0 & -\frac{1}{6} & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 4 & 0 \\ 0 & -18 & -36 & 7 \\ 0 & 0 & 6 & -\frac{4}{9} \\ 0 & 0 & 0 & \frac{49}{18} \end{bmatrix}$$

This is the $A = LU$ factorization, but let's use the other method to double-check our work, starting with setting up $A = LU$.

$$\begin{bmatrix} 1 & 3 & 4 & 0 \\ 9 & 9 & 0 & 7 \\ 7 & 7 & 6 & 5 \\ 0 & 3 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ & 1 & 0 & 0 \\ & & 1 & 0 \\ & & & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 4 & 0 \\ 9 & 9 & 0 & 7 \\ 7 & 7 & 6 & 5 \\ 0 & 3 & 0 & 2 \end{bmatrix}$$

We only need to zero-out the entries in U that are below the main diagonal. To zero-out the 9 in $U_{2,1}$, we need to subtract 9 of R_1 from R_2. Since that row operation is $R_2 - 9R_1$, we put 9 into $L_{2,1}$.

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$$\begin{bmatrix} 1 & 3 & 4 & 0 \\ 9 & 9 & 0 & 7 \\ 7 & 7 & 6 & 5 \\ 0 & 3 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 9 & 1 & 0 & 0 \\ & & 1 & 0 \\ & & & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 4 & 0 \\ 0 & -18 & -36 & 7 \\ 7 & 7 & 6 & 5 \\ 0 & 3 & 0 & 2 \end{bmatrix}$$

To zero-out the 7 in $U_{3,1}$, we need to subtract 7 of R_1 from R_3. Since the row operation is $R_3 - 7R_1$, we put 7 into $L_{3,1}$.

$$\begin{bmatrix} 1 & 3 & 4 & 0 \\ 9 & 9 & 0 & 7 \\ 7 & 7 & 6 & 5 \\ 0 & 3 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 9 & 1 & 0 & 0 \\ 7 & & 1 & 0 \\ & & & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 4 & 0 \\ 0 & -18 & -36 & 7 \\ 0 & -14 & -22 & 5 \\ 0 & 3 & 0 & 2 \end{bmatrix}$$

To zero-out the -14 in $U_{3,2}$, we need to subtract $14/18 = 7/9$ of R_2 from R_3. Since the row operation is $R_3 - (7/9)R_2$, we put $7/9$ into $L_{3,2}$.

$$\begin{bmatrix} 1 & 3 & 4 & 0 \\ 9 & 9 & 0 & 7 \\ 7 & 7 & 6 & 5 \\ 0 & 3 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 9 & 1 & 0 & 0 \\ 7 & \frac{7}{9} & 1 & 0 \\ & & & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 4 & 0 \\ 0 & -18 & -36 & 7 \\ 0 & 0 & 6 & -\frac{4}{9} \\ 0 & 3 & 0 & 2 \end{bmatrix}$$

We don't need to perform any row operation to keep the 0 in $L_{4,1}$, so

$$\begin{bmatrix} 1 & 3 & 4 & 0 \\ 9 & 9 & 0 & 7 \\ 7 & 7 & 6 & 5 \\ 0 & 3 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 9 & 1 & 0 & 0 \\ 7 & \frac{7}{9} & 1 & 0 \\ 0 & & & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 4 & 0 \\ 0 & -18 & -36 & 7 \\ 0 & 0 & 6 & -\frac{4}{9} \\ 0 & 3 & 0 & 2 \end{bmatrix}$$

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To zero-out the 3 in $U_{4,2}$, we need to add $3/18 = 1/6$ of R_2 to R_4. The row operation is $R_4 + (1/6)R_2$, but we want to see subtraction in the row operation, so we write it as $R_4 - (-1/6)R_2$, then we put $-1/6$ into $L_{4,2}$.

$$\begin{bmatrix} 1 & 3 & 4 & 0 \\ 9 & 9 & 0 & 7 \\ 7 & 7 & 6 & 5 \\ 0 & 3 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 9 & 1 & 0 & 0 \\ 7 & \frac{7}{9} & 1 & 0 \\ 0 & -\frac{1}{6} & & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 4 & 0 \\ 0 & -18 & -36 & 7 \\ 0 & 0 & 6 & -\frac{4}{9} \\ 0 & 0 & -6 & \frac{19}{6} \end{bmatrix}$$

To zero-out the -6 in $U_{4,3}$, we need to add 1 of R_3 to R_4. The row operation is $R_4 + 1R_3$, but we want to see subtraction in the row operation, so we write it as $R_4 - (-1)R_3$, then we put -1 into $L_{4,3}$.

$$\begin{bmatrix} 1 & 3 & 4 & 0 \\ 9 & 9 & 0 & 7 \\ 7 & 7 & 6 & 5 \\ 0 & 3 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 9 & 1 & 0 & 0 \\ 7 & \frac{7}{9} & 1 & 0 \\ 0 & -\frac{1}{6} & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 4 & 0 \\ 0 & -18 & -36 & 7 \\ 0 & 0 & 6 & -\frac{4}{9} \\ 0 & 0 & 0 & \frac{49}{18} \end{bmatrix}$$

Using both methods, we've arrived at the same $A = LU$ factorization.

Factoring out the pivots

Decomposing A into LU gives us the lower triangular matrix L with all 1s along the main diagonal, and the upper triangular matrix U. But sometimes we'll want to rewrite U so that it also has all 1s along the main diagonal.

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Let's take the 3×3 matrix we looked at earlier. We had already factored it into $A = LU$ as

$$\begin{bmatrix} 5 & 1 & 3 \\ 4 & 2 & 5 \\ 8 & 6 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{4}{5} & 1 & 0 \\ \frac{8}{5} & \frac{22}{6} & 1 \end{bmatrix} \begin{bmatrix} 5 & 1 & 3 \\ 0 & \frac{6}{5} & \frac{13}{5} \\ 0 & 0 & -\frac{31}{3} \end{bmatrix}$$

To change the entries along the main diagonal of U, we start by inserting an identity matrix into the equation, changing it to $A = LIU$. This doesn't change the value of the equation at all, because multiplying by the identity matrix is like multiplying by 1.

$$\begin{bmatrix} 5 & 1 & 3 \\ 4 & 2 & 5 \\ 8 & 6 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{4}{5} & 1 & 0 \\ \frac{8}{5} & \frac{22}{6} & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 5 & 1 & 3 \\ 0 & \frac{6}{5} & \frac{13}{5} \\ 0 & 0 & -\frac{31}{3} \end{bmatrix}$$

We want to change the entry in $U_{1,1}$ from a 5 to a 1, which we can do by dividing through the first row of U by 5, and putting the 5 into the identity matrix.

$$\begin{bmatrix} 5 & 1 & 3 \\ 4 & 2 & 5 \\ 8 & 6 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{4}{5} & 1 & 0 \\ \frac{8}{5} & \frac{22}{6} & 1 \end{bmatrix} \begin{bmatrix} 5 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & \frac{1}{5} & \frac{3}{5} \\ 0 & \frac{6}{5} & \frac{13}{5} \\ 0 & 0 & -\frac{31}{3} \end{bmatrix}$$

To change the entry in $U_{2,2}$ from a $6/5$ to a 1, we'll divide through the second row of U by $6/5$, putting the $6/5$ into the identity matrix.

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$$\begin{bmatrix} 5 & 1 & 3 \\ 4 & 2 & 5 \\ 8 & 6 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{4}{5} & 1 & 0 \\ \frac{8}{5} & \frac{22}{6} & 1 \end{bmatrix} \begin{bmatrix} 5 & 0 & 0 \\ 0 & \frac{6}{5} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & \frac{1}{5} & \frac{3}{5} \\ 0 & 1 & \frac{13}{6} \\ 0 & 0 & -\frac{31}{3} \end{bmatrix}$$

Then finally, to change the entry in $U_{3,3}$ from a $-31/3$ to a 1, we'll divide through the third row of U by $-31/3$, putting the $-31/3$ into the identity matrix.

$$\begin{bmatrix} 5 & 1 & 3 \\ 4 & 2 & 5 \\ 8 & 6 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{4}{5} & 1 & 0 \\ \frac{8}{5} & \frac{22}{6} & 1 \end{bmatrix} \begin{bmatrix} 5 & 0 & 0 \\ 0 & \frac{6}{5} & 0 \\ 0 & 0 & -\frac{31}{3} \end{bmatrix} \begin{bmatrix} 1 & \frac{1}{5} & \frac{3}{5} \\ 0 & 1 & \frac{13}{6} \\ 0 & 0 & 1 \end{bmatrix}$$

Let's continue with the 4×4 matrix example from earlier, so that we can see one more time how to change the entries along the main diagonal of U.

Example

Rewrite the $A = LU$ factorization so that U has only 1s along its main diagonal.

$$\begin{bmatrix} 1 & 3 & 4 & 0 \\ 9 & 9 & 0 & 7 \\ 7 & 7 & 6 & 5 \\ 0 & 3 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 9 & 1 & 0 & 0 \\ 7 & \frac{7}{9} & 1 & 0 \\ 0 & -\frac{1}{6} & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 4 & 0 \\ 0 & -18 & -36 & 7 \\ 0 & 0 & 6 & -\frac{4}{9} \\ 0 & 0 & 0 & \frac{49}{18} \end{bmatrix}$$

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Let's start by inserting an identity matrix into the equation.

$$\begin{bmatrix} 1 & 3 & 4 & 0 \\ 9 & 9 & 0 & 7 \\ 7 & 7 & 6 & 5 \\ 0 & 3 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 9 & 1 & 0 & 0 \\ 7 & \frac{7}{9} & 1 & 0 \\ 0 & -\frac{1}{6} & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 4 & 0 \\ 0 & -18 & -36 & 7 \\ 0 & 0 & 6 & -\frac{4}{9} \\ 0 & 0 & 0 & \frac{49}{18} \end{bmatrix}$$

We'll first pull out the -18 from $U_{2,2}$, dividing the second row of U by -18.

$$\begin{bmatrix} 1 & 3 & 4 & 0 \\ 9 & 9 & 0 & 7 \\ 7 & 7 & 6 & 5 \\ 0 & 3 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 9 & 1 & 0 & 0 \\ 7 & \frac{7}{9} & 1 & 0 \\ 0 & -\frac{1}{6} & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -18 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 4 & 0 \\ 0 & 1 & 2 & -\frac{7}{18} \\ 0 & 0 & 6 & -\frac{4}{9} \\ 0 & 0 & 0 & \frac{49}{18} \end{bmatrix}$$

Then we'll pull out the 6 from $U_{3,3}$, dividing the third row of U by 6.

$$\begin{bmatrix} 1 & 3 & 4 & 0 \\ 9 & 9 & 0 & 7 \\ 7 & 7 & 6 & 5 \\ 0 & 3 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 9 & 1 & 0 & 0 \\ 7 & \frac{7}{9} & 1 & 0 \\ 0 & -\frac{1}{6} & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -18 & 0 & 0 \\ 0 & 0 & 6 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 4 & 0 \\ 0 & 1 & 2 & -\frac{7}{18} \\ 0 & 0 & 1 & -\frac{2}{27} \\ 0 & 0 & 0 & \frac{49}{18} \end{bmatrix}$$

Finally, we'll pull out the $49/18$ from $U_{4,4}$, dividing the fourth row of U by $49/18$.

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$$\begin{bmatrix} 1 & 3 & 4 & 0 \\ 9 & 9 & 0 & 7 \\ 7 & 7 & 6 & 5 \\ 0 & 3 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 9 & 1 & 0 & 0 \\ 7 & \frac{7}{9} & 1 & 0 \\ 0 & -\frac{1}{6} & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -18 & 0 & 0 \\ 0 & 0 & 6 & 0 \\ 0 & 0 & 0 & \frac{49}{18} \end{bmatrix} \begin{bmatrix} 1 & 3 & 4 & 0 \\ 0 & 1 & 2 & -\frac{7}{18} \\ 0 & 0 & 1 & -\frac{2}{27} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Now L is a lower triangular matrix, U is an upper triangular matrix, and both matrices have only 1s along their main diagonals.

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